

Kiran Pradeep

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[LinkedIn](#) | [GitHub](#) | [Google Scholar](#) | [Website](#)

EDUCATION

- Indian Institute of Technology Bombay (IITB)** Aug 2023 – May 2026
M.S. by Research in Computer Science & Engineering
(Advisor: Prof. [Pushpak Bhattacharyya](#) & Prof. [Sunita Sarawagi](#)) CPI: 9.03
- Kirori Mal College, Delhi University** Jul 2019 – Jun 2021
M.Sc. in Mathematics CPI: 8.07
- Sri Venkateswara College, Delhi University** Jul 2015 – Jun 2018
B.Sc. (Hons.) in Mathematics CPI: 6.67

PROFESSIONAL EXPERIENCE

- Research Extern, IBM Research** (May 2024 – Aug 2024)
(Mentors: [Nishtha Madaan](#), [Sameep Mehta](#))
 - Engineered a plug-and-play NL-to-SQL multi-agentic system utilizing **open-source LLMs** (LLaMA-3.1-70B, phi-4) without fine-tuning, outperforming GPT-4-based baselines in cost-efficiency and scalability.
 - Pioneered question decomposition at the schema linking stage, boosting column recall by **25.1%** and execution accuracy by **8.2%** on the BIRD benchmark.
- AI Software Development Engineer, Asmadiya Technologies Pvt. Ltd.** Dec 2021 – Jul 2023
 - Architected the backend infrastructure for a scalable ML model training platform using **FastAPI**, enabling distributed dataset processing and custom image classification model deployment.
 - Automated high-throughput ML workflows using **AWS Lambda**, **Terraform** EC2 provisioning, and **Docker** containers, optimizing cluster reliability for scheduled training pipelines.
 - Deployed a highly accurate (**93%**) object detection system using YOLOv5, reducing false positives by 50%.

PUBLICATIONS

- Divide, Link, and Conquer: Recall-oriented Schema Linking for NL-to-SQL via Question Decomposition.**
Kiran Pradeep, Kirushikesh DB, Nishtha Madaan, Sameep Mehta, Pushpak Bhattacharyya. *Proceedings of the 2025 Empirical Methods in Natural Language Processing (EMNLP): Industry Track, Suzhou, China, November 5-9, 2025.*
- RG-VQA: Leveraging Retriever-Generator Pipelines for Knowledge Intensive Visual Question Answering.**
Settaluri Lakshmi Sravanthi, Pulkit Agarwal, Debjyoti Mondal, Rituraj Singh, Subhadarshi Panda, Ankit Mishra, **Kiran Pradeep**, Srihari K B, Godawari Sudhakar Rao, Pushpak Bhattacharyya. *Findings of the Association for Computational Linguistics: EMNLP, Suzhou, China, November 5-9, 2025.*

RESEARCH PROJECTS

- End-to-End Knowledge-Enhanced Visual Question Answering** (Jan'24–Apr'24)
(MS R&D - II; Guide: Prof. [Pushpak Bhattacharyya](#))
 - Trained a retriever for the VQA task without explicit supervision using the loss signals from the reader.
 - Attained **68.5%** test accuracy on A-OKVQA using **ConceptNet** as the external knowledge, surpassing a 61.4% baseline by jointly training retriever and reader with architecture modifications.
- Diffusion Models and Classifier-Free Guidance** (Jan'25–Apr'25)
(CS726 - Advanced Machine Learning | Instructor: Prof. [Sunita Sarawagi](#))
 - Investigated impact of **diffusion steps** and **noise schedules** on generative quality using metrics like **Earth Mover Distance (EMD)** and **Negative Log-Likelihood (NLL)** across 2D datasets.
 - Implemented and evaluated **Classifier-Free Guidance** to improve conditional sample quality, achieving significant generation gains in **accuracy** and **F1-score** at moderate guidance scales.

TECHNICAL SKILLS

- Core Domains:** NLP, Computer Vision, Large Language Models (LLMs), Transformer, Reinforcement Learning
- Languages & Frameworks:** Python, PyTorch, Hugging Face, LangChain, FastAPI, TensorFlow
- Infrastructure & DevOps:** Slurm, Conda, Docker, Git, AWS, Terraform, MongoDB

ACHIEVEMENTS

- Awarded the **Thomas Dooie Class of 1974 Award** by the CSE Department, IIT Bombay, for outstanding MS applicant during M.S. admissions (May 2023), received a cash prize of Rs. 1,00,000.
- Secured **AIR 744** in GATE Computer Science 2023 amongst 75,680 appeared candidates.
- Secured **AIR 598** in IIT JAM - Mathematics in 2019 amongst 12,863 appeared candidates.